SUPPORTING ADULT GUIDANCE

Both sides, and brackets

Equations, Inequalities and Simultaneous Relationships · Lesson 2. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This lesson extends equation-solving to two harder forms: the unknown on **both sides**, and equations with **brackets**. The organising idea is 'tidy before you solve' — collect like terms and expand brackets to reduce each equation to the balance problem of EQ-01. Checking by substitution remains the non-negotiable final step.

Driving Question: *What if the unknown is on both sides, or hidden inside brackets?*

The mathematics, in plain terms

When x appears on both sides, subtracting the smaller x-term from both sides preserves the equality and gathers the unknown on one side. A bracket a(x + b) is shorthand for a rectangle a by (x + b); expanding — a(x + b) = ax + ab — distributes the multiplier across every term inside. After either tidying step the equation is a two-step linear equation solved by inverse operations, and substitution verifies the solution.

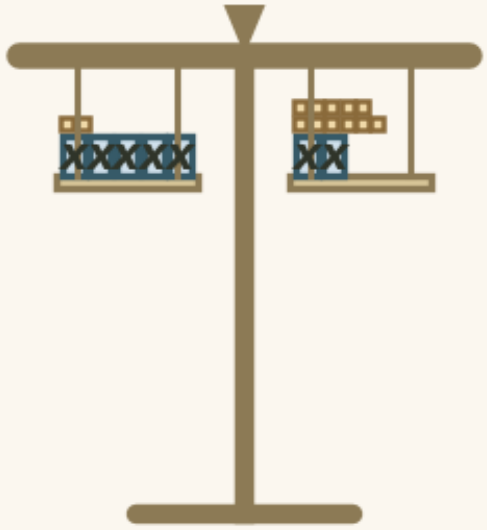
The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — The unknown on both sides

Solve $5x + 2 = 2x + 11$ — there are x's on both sides.

$$5x + 2 = 2x + 11$$

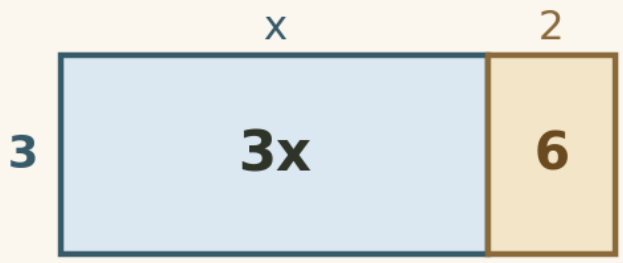


1. Collect x: subtract $2x$ from both sides.	$5x + 2 - 2x = 2x + 11$ $- 2x$
2. The x is now on one side only.	$3x + 2 = 11$
3. Subtract 2 from both sides.	$3x = 9$
4. Divide both sides by 3.	$x = 3$
5. Check in the original.	$5\times3 + 2 = 2\times3 + 11$ $= 17$ ✓

$x = 3$ — subtract $2x$ to collect the unknown, then solve $3x + 2 = 11$; the check gives 17 on both sides.

Example 2 — A bracket to expand

Solve $3(x + 2) = 18$ — the x is inside a bracket.



1.	Expand: the 3 multiplies both x and 2.	$3(x + 2) = 3x + 6$
2.	Rewrite the equation.	$3x + 6 = 18$
3.	Subtract 6 from both sides.	$3x = 12$
4.	Divide both sides by 3.	$x = 4$
5.	Check in the original.	$3(4 + 2) = 18$ ✓

$x = 4$ — expand to $3x + 6 = 18$, then solve; the check $3(4 + 2) = 18$ confirms it.

Tidy first, then keep the balance

Encourage a fixed order: expand any brackets, then collect the unknown onto one side, then clear the number, then divide, then check. The two common traps are expanding only the first term inside a bracket, and dividing while x is still on both sides. A learner who tidies first meets only equations they can already solve.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Collect a jumble of algebra tiles into ax + b — like with like.	~8 min
Movement	Solve on a two-sided balance: remove the same bags from both pans to collect x.	~13 min
Reflection	See collecting and solving written down, line by line, then check.	~10 min
Creativity	Expand a bracket with an area model, then use it to solve.	~9 min
Rest	A short pause on the idea of tidying without changing meaning.	~2 min

Nutrition	Name what the learning feeds: a tidy mind, simplifying before solving.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Expanding only the first term: $3(x + 2) = 3x + 2$.
Return to the rectangle: 'The height 3 covers the whole width — both the x part and the 2 part. What is 3 lots of 2?' So $3x + 6$.

Subtracting x from only one side when collecting.
Same rule as always: 'What you do to one side you do to the other. You subtracted $2x$ on the left — what happens on the right?'

Collecting the larger x-term and getting a negative coefficient.
Suggest subtracting the smaller x-term so the unknown stays positive: 'Which side has fewer x's? Take that many from both.'

Solving but never checking.
Make substitution routine: 'Put your value back into the original — do both sides come out equal?'

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson's Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: collecting like terms, expanding brackets, and keeping an equation balanced (EQ-01). A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner writes $3(x + 2) = 3x + 2$, or subtracts $2x$ from one side only. The aim is a question that hands the reasoning back.

- "In $3(x + 2)$, what does the 3 multiply — just the x, or everything inside?"

- “You took $2x$ off the left to collect the x's — what must happen on the right?”
- “Which side has fewer x's? Subtract that many from both, so the x stays positive.”
- “Put your answer back into the original: do both sides come out the same?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Expand the bracket, or divide first?

Both are valid. Expanding is the most general habit and always works; dividing first is neat when the outside number divides the other side cleanly (e.g. $2(x + 3) = 16 \rightarrow x + 3 = 8$). Let the learner choose and justify.

What if collecting gives a negative number of x?

It still works, but positives are easier early on. Subtracting the smaller x-term from both sides keeps the coefficient positive.

Do the numbers always come out whole?

Not always — the stretch question gives $x = 2.5$. Fractions and decimals are perfectly good solutions; the check works the same way.

How does this build on Lesson 1?

Every equation here becomes an EQ-01 equation once tidied. The new skills are the tidying steps — collecting like terms and expanding brackets — not a new way of solving.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.

Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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